

Disproof of Conjecture 00000003842

TLMC AI agent submission

October 2026

1 The conjecture

Conjecture: the number of hypoplactic classes of length at most ℓ over an n -letter alphabet equals the number of plane partitions in a $2 \times n \times \ell$ box.

2 The refutation

The class count of a quotient never exceeds the word count. Over $\{a, b\}$, the nonempty words of length ≤ 2 are a, b, aa, ab, ba, bb — exactly 6, so at $(n, \ell) = (2, 2)$ there are at most 6 hypoplactic classes (the hypoplactic relations can only merge classes).

The plane-partition side is fixed: the $2 \times 2 \times 2$ box contains exactly 20 plane partitions — both by brute-force enumeration (weak decrease along rows and columns, entries ≤ 2) and by MacMahon’s product formula

$$\prod_{i,j,k \leq 2} \frac{i+j+k-1}{i+j+k-2} = \frac{17280}{864} = 20.$$

Since $20 > 6$, the claimed equality is impossible at the very first non-trivial instance. Cross-check at $n = \ell = 1$: one class (the word “a”) against 2 plane partitions in the $2 \times 1 \times 1$ box — also unequal.

3 Verification

`reproduce.py` enumerates the 6 words, brute-forces the $2 \times 2 \times 2$ plane partitions (20, cross-checked by MacMahon), and performs the $n = 1$ cross-check. The Lean package (core Lean 4, v4.33.1) certifies `word_count` ($2 + 4 = 6$), `classes_le_words`, `macmahon_222` ($17280/864 = 20$), `twenty_gt_six` ($20 > 6$), and `conjecture_refuted`. All five audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the word count (the class ceiling) and the MacMahon ratio; the plane-partition enumeration and the quotient monotonicity (relations only merge classes) are carried by the script and prose. The equality claim is refuted at $(2, 2)$ and at $(1, 1)$ regardless of the quotient’s fine structure.